University of Groningen
Exam Introduction to Econometrics
Monday, June 20 2013, 6:30 PM – 9:30 PM

1. 70 points

Consider a sample of 4744 working individuals, born in 1958 and living in Great Britain. We know their hourly wage ($wage$) in 1991, the number of years of education ($educ$), the number of years of experience ($exper$). We also know their mathematical ability ($math$) and their verbal ability ($verbal$) in 1965. Let us define the variable $ability = math + verbal$, the variable $lwage = log(wage)$ and the variable $educ2 = educ^2$.

Consider the following OLS regression:

$$lwage_i = \beta_1 + \beta_2 educ_i + \beta_3 educ2_i + \beta_4 exper_i + \beta_5 ability_i + \beta_6 math_i + \beta_7 SW_i + \beta_8 SE_i + \beta_9 female_i + \varepsilon_i \quad (1)$$

where $female_i$ is a dummy equal to 1 if the person is female. In order to account for regional differences in wages, we have also added two regional dummies to equation (1): The dummy variable SE_i (SW_i) indicates that the respondent lives in the South East (Sooth West) part of Great Britain[1]. The results of regression (1) are presented in Table 1 (cf. column 'equation 1').

(a) Provide a careful economic interpretation of the estimated coefficient on SW_i.

(b) How large is the return to education for a worker with 10 years of education ($educ_i = 10$)? And for a worker with 20 years of education?

(c) Use a t test to check the validity of the null hypothesis that mathematical ability and verbal ability have the same effect on $lwage_i$. Please mention all steps of the testing procedure ($\alpha = 0.05$).

(d) Table 1 also presents the results of a regression model without the regional dummies (cf. column 'equation 2'). Check by means of a F test whether the regional dummies are jointly significant. Please mention all steps of the testing procedure ($\alpha = 0.05$).

(e) Now, we consider the possibility that the β coefficients differ across subgroups of males and females. We have estimated the following model for both subgroups (cf. columns (`equation male`) and (`equation female`) in Table 1):

$$lwage_i = \beta_1^k + \beta_2^k educ_i + \beta_3^k educ2_i + \beta_4^k exper_i + \beta_5^k ability_i + \beta_6^k math_i + \beta_7^k SW_i + \beta_8^k SE_i + \varepsilon_i, \; k = \text{male, female} \quad (2)$$

Test the null hypothesis that that all *slope* coefficients of model (2) are the same across the subgroups of "females" and "males". Notice that according to the null hypothesis the intercept term β_1 might differ across the two subgroups ($\alpha = 0.05$).

(f) Up to now we have assumed that the error term is homoskedastic. In order to check the validity of this assumption we have run the following auxiliary regression (see column 'aux. regression 1' in Table 1):

$$e_i^2 = \gamma_1 + \gamma_2 educ_i + \gamma_3 educ2_i + \gamma_4 exper_i + \gamma_5 ability_i + \gamma_6 math_i + \gamma_7 SW_i + \gamma_8 SE_i + \gamma_9 female_i + u_i \quad (3)$$

where e_i is the residual of the estimated equation (1). Please carry out a test to show that the error term of equation (1) is heteroskedastic ($\alpha = 0.05$).

(g) In light of the results of the heteroskedasticity test above, we have to conclude that the results of the Chow test which you obtained in part (1e) of this exercise, cannot be trusted. However, this test can be carried out in an alternative way which takes into account the heteroskedastic nature of the error term. Please indicate precisely how this alternative Chow test can be carried out. Write down the regression model which you need to estimate in order to carry out this alternative test.

(h) Now consider the following simplified version of regression equation (1)

$$lwage_i = \beta_1 + \beta_2 educ_i + \beta_4 exper_i + \beta_7 SW_i + \beta_8 SE_i + \beta_9 female_i + u_i \quad (4)$$

Why might $educ$ be endogenous?

[1] Reference group consists of respondents living in the Northern part of Great Britain.

(i) We treat $educ_i$ as an endogenous right hand side variable in equation (4). The reduced form demand for the education equation ("first stage equation") is specified as follows:

$$educ_i = \delta_1 + \delta_2 exper_i + \delta_3 SW_i + \delta_4 SE_i + \delta_5 female_i + \delta_6 payrsed_i + \delta_8 mayrsed_i + \upsilon_i \qquad (5)$$

where $payrsed_i$ and $mayrsed_i$ denote father's and mother's number of years of education respectively. The parameters of equation (4) are estimated by means of 2SLS (cf. column '2sls' of Table 1). Table 1 also presents the estimation results of equation (5) (see column 'first stage reg').

Under which conditions will the 2SLS estimator yield consistent estimators of the β parameters in equation (4)? Do you think that these conditions are satisfied?

(j) Use the information presented in Table 1 to test the hypothesis that in equation (4) the right hand side variable $educ_i$ is exogenous ($\alpha = 0.05$).

(k) Use the information presented in Table 1 to perform the Sargan test of overidentifying restriction(s) ($\alpha = 0.05$). What is the underlying null hypothesis of the Sargan test?

ANSWERS EXERCISE 1

(a) Keeping other factors constant, people living in the South Wrest earn 21.9 % (precise calculation (exp(0.219)-1)*100=24.48 %) more than the reference group, i.e. people who live in the Northern part of Great Britain.

(b) Marginal effect at educ=10: $b_{educ} + 2 \cdot b_{educ2} \cdot 10 = .3834481 + 2 * (-.0097959) * 10 = .1875301$

Marginal effect at educ=20: $b_{educ} + 2 \cdot b_{educ2} \cdot 20 = .3834481 + 2 * (-.0097959) * 20 = -.0083879$

(c) Equation (1) cab be rewritten as follows:

$$lwage_i = \beta_1 + \beta_2 educ_i + \beta_3 educ2_i + \beta_4 exper_i + \beta_5 (math_i + verbal_i) + \beta_6 math_i + \beta_7 SW_i + \beta_8 SE_i + \beta_9 female_i + \varepsilon_i \qquad (6)$$

So we like to test the following $H_0 : \beta_6 = 0$ (then $verbal_i$ and $math_i$ have the same effect, β_5, on $lwage_i$)

Steps testing procedure

- Formulate H_0: $\beta_6 = 0$, $H_1 : \beta_6 \neq 0$
- $\alpha = 0.05$
- Test statistic $\frac{b_6}{se(b_6)}$. This test statistic follows a student t distribution with $n - K$ degrees of freedom. $n = 4744$ and $K = 9$.
- Compute the value of the t statistic. Notice that $t = 0.00372/0.00758 = .49076517$
- Look up critical value ($\alpha = 0.05, n - K = 4735$): 1.96
- Since $|0.490| < 1.960$ we do not reject the H_0. We find evidence that mathematical ability and verbal ability have the same effect on the hourly wage rate.

(d) Steps F-test

- Formulate H_0 and H_1
  $H_0 : \beta_7 = 0, \beta_8 = 0$, $H_1 : H_0$ not true
- Test statistic
  $$F(\#r, n - K) = \frac{(SSR_r - SSR_u)/\#r}{SSR_u/(n - K)}$$
  This test statistic follows a student F distribution with $\#r$ numerator degrees of freedom and $n - K$ denominator degrees of freedom.
  In this case, $n = 4744, K = 9, \#r = 2$,
- Compute the value of the F statistic. Notice that $SSR_r = 670.794638, SSR_u = 643.600006$
  $$F(2, 4735) = \frac{(670.794638 - 643.600006)/2}{643.600006/(4735)} = 100.03619$$
- Look up critical value ($\alpha = 0.05, \#r = 2, n - K = 4735$): 3.000

- Since 100.03619 > 3.000. So the regional dummies SE and SW are jointly statistically significant at the 5 % level.

(e) Steps F test

- Formulate H_0

$$H_0 : \beta_i^{female} = \beta_i^{male}, i = 2, \ldots, 8$$

Apart from the intercept term, the coefficients of the "female equation" are the same as those of the "male equation".
Number of restrictions imposed by H_0: $\#r = 7$

- Computation F statistic

$$F(\#r, n-K) = \frac{(SSR_r - SSR_u)/\#r}{SSR_u/(n-K)}$$

where K = number of regression parameters in unrestricted model
Unrestricted model (separate equations for female=1 and female=0)
Number of regression parameters in unrestricted model: $2 \times 8 = 16$
SSR unrestricted model: $SSR_u = SSR^{female} + SSR^{male} = 323.880 + 315.191 = 639.071$
SSR restricted model: $SSR_r = 643.600$ (see column 'equation 1 table 1)
So, The value of the F-statistic equal to:

$$F(7, 4744-16) = \frac{(643.600 - 639.071)/7}{639.071/(4744-16)} = 4.7866606$$

- Look up critical value: $c = 2.010$
- $4.7866606 > 2.010$, we reject the H_0. We conclude that (at least one of the) slope coefficients are different across the subgroups of males and females.

(f) Steps heteroskedasticity test

- Formulate H_0 (error term is homoskedastic.)

$$H_0 : \gamma_2 = \gamma_3 = \ldots = \gamma_9 = 0 \text{ or } H_0 : E(\varepsilon_i^2 | \boldsymbol{x}_i) = \sigma^2 \text{ (constant variance)}$$

.
Number of restrictions imposed by H_0: $\#r = 8$

- Computation χ^2 statistic

$$\chi^2(8) = n \cdot R^2 = 4744 * 0.007 = 33.208$$

- Look up critical value $\chi^2(8)$: $c = 15,50731$
- $33.208 > 15.50731$, we reject the H_0. We conclude that the error term is heteroskedastic (the variance of the error term varies with background characteristics).

(g) Estimate the following equation by means of OLS and compute the heteroskedasticity-robust standard errors:

$$\begin{gathered} lwage_i = \beta_1 + \beta_2 educ_i + \beta_3 educ2_i + \beta_4 exper_i + \beta_5 ability_i + \beta_6 math_i + \beta_7 SW_i + \beta_8 SE_i + \beta_9 female_i + \\ \beta_{10} female_i * educ_i + \beta_{11} educ2_i + \beta_12 female_i * exper_i + \beta_{13} female_i * ability_i + \\ \beta_{14} female_i * math_i + \beta_{15} female_i * SW_i + \beta_{16} female_i * SE_i + \varepsilon_i \end{gathered} \tag{7}$$

Next test the following H_0: $H_0 : \beta_{10} = \ldots = \beta_{16} = 0$ or $H_0 : \boldsymbol{Rb} = \boldsymbol{0}$ where the $((7 \times 16)$ matrix $\boldsymbol{R}$ is defined as follows: $\boldsymbol{R} = (\boldsymbol{O}, \boldsymbol{I}_7)$ where $\boldsymbol{O}$ is a (7×9) matrix containing zero elements and $\boldsymbol{I}_7$ is the Identity matrix of order 7.

The associated test statistic (Wald statistic) is equal to (robust option in Stata):

$$W \equiv n(\boldsymbol{Rb})'[\boldsymbol{R}[\widehat{Avar(\boldsymbol{b})}]\boldsymbol{R}']^{-1}(\boldsymbol{Rb}) \underset{d}{\rightarrow} \chi^2(7) \tag{8}$$

where the estimated heteroskedasticity-robust covariance matrix of $\sqrt{n}(\boldsymbol{b}-\boldsymbol{\beta})$ ($\widehat{Avar(b)}$) is equal to

$$\widehat{Avar(b)} = \boldsymbol{S}_{xx}^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i \boldsymbol{x}_i'\right)\boldsymbol{S}_{xx}^{-1} \tag{9}$$

($\boldsymbol{S}_{xx} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{x}_i'$)

(h) Compared with Model (1), model (4) misses the following regressors: $educ_i^2$, $math_i$ and $ability_i$. These three variables are obviously positively correlated with $educ_i$. Since $u_i = \varepsilon_i + \beta_3 educ2_i + \beta_5 ability_i + \beta_6 math_i$, $educ_i$ is an endogenous rhs variable, i.e. correlated with the error term u_i of equation (4).

(i) Assumptions

3.1: Linearity Model (4) is in indeed linear in the parameters. This assumption does not need to be mentioned

3.2': random sample It is reasonable to assume that we have a random sample available. This assumption does not need to be mentioned

3.4: rank condition (instrument relevance) The matrix $E(\boldsymbol{x}_i \boldsymbol{z}_i')$ should be of full column rank where $\boldsymbol{x}_i = (1, exper_i, SE_i, SW_i, female_i, payrsed_i, mayrsed_i)'$ and $\boldsymbol{z}_i = (1, educ_i, exper_i, SE_i, SW_i, female_i)'$. **This condition is likely to be satisfied** because the endogenous rhs variable is likely to be partially correlated with the 'excluded instruments' $payrsed_i$ and $mayrsed_i$. The results from the first regression suggest that the two excluded instruments $payrsed_i$ and $mayrsed_i$ are strongly jointly significant: The F statistic takes on a large value (F=46.736). This number is larger than 10, which gives us confidence that the instrument relevance condition is met.

3.3: Instrument exogeneity $E(\boldsymbol{x}_i u_i) = \boldsymbol{0}$. **This assumption is definitely not satisfied**. The variables $payrsed_i$ and $mayrsed_i$ are definitely not orthogonal to $u_i = \varepsilon_i + \beta_3 educ2_i + \beta_5 ability_i + \beta_6 math_i$ because $payrsed_i$ and $mayrsed_i$ are certain correlated with $educ_i^2$ because we have shown above that these 2 variables are partially correlated with $educ_i$. By omitting the variable $educ_i^2$ from the specification, it is impossible to find suitable instruments. Notice also that the two excluded instruments are also likely to be partially correlated with $math_i$ and $ability_i$.

(j) We just concluded that $payrsed_i$ and $mayrsed_i$ are no valid instruments. Strictly speaking it does not make sense to carry out the exogeneity test and the Sargan test. But the sake of it we carry out these tests

Exogeneity test: $H_0 : E(educ_i u_i) = 0$

Hausman-Wu test of exogeneity can be carried by running the following auxiliary regression:

$$lwage_i = \beta_1 + \beta_2 educ_i + \beta_4 exper_i + \beta_7 SW_i + \beta_8 SE_i + \beta_9 female_i + \theta \hat{\upsilon}_i + \epsilon_i \tag{10}$$

where $\hat{\upsilon}_i$ denotes the residual of the first stage regression. The test statistic is equal to the t-value of the estimated parameter associated wit the first stage residual (see Table 1 column aux regression 2).

$$t_{\hat{\upsilon}} = \frac{-0.0372}{0.0219} = -1.6986301$$

Since $|-1.6986301| < 1.96$ ($\alpha = 0.05$), the H_0 of exogeneity is not rejected. We do not find strong evidence that the right hand side variable $educ_i$ is endogenous.

(k) $H_0 : E(\boldsymbol{x}_i u_i) = \boldsymbol{0}$, i.e. all instruments (the predetermined regressors and the excluded instruments) should be orthogonal to the error term u_i ($\boldsymbol{x}_i = (1, exper_i, SE_i, SW_i, female_i, payrsed_i, mayrsed_i)'$)

The Sargan test can be carried out by running an auxiliary regression of the second stage residual $\hat{u}_{lwage}$ on all instruments (i.e predetermined regressors plus the excluded instruments), see the last column of table 1. The Sargan statistic is equal to

$$n \times R^2 = 4744 * 0.0004 = 1.8976$$

where R^2 is the R-squared of the regression mentioned above. The asymptotic distribution of this test statistic is $\chi^2(K-L)$, where K the number of instruments and L the number of regressors in equation (4). In our case $K = 7$ and $L = 6$ $(K - L = 1)$. So, The critical value $(\alpha = 0.05)$ is equal to 3,84.

Since $1.8976 < 3.84$, we do not reject H_0. Conclusion: The H_0 that the instruments are orthogonal to the error term of eq. (4) is not rejected

2. 16 points

a) Let $u = y - E(y|\boldsymbol{x})$. Show that $E[g(\boldsymbol{x})u] = 0$ for any function $g(\boldsymbol{x})$.
HINT: $E[g(\boldsymbol{x})u] = E(E[g(\boldsymbol{x})u|\boldsymbol{x}])$

b) Show that

$$var(y) = E[var(y|\boldsymbol{x})] + var\left(E(y|\boldsymbol{x})\right) \tag{11}$$

HINT 1: The conditional variance of a random scalar y given a random vector $\boldsymbol{x}$ is defined as

$$var(y|\boldsymbol{x}) = \sigma^2(\boldsymbol{x}) = E[(y - E(y|\boldsymbol{x}))^2|\boldsymbol{x}] \tag{12}$$

HINT 2: Use the add and subtract strategy. $y - E(y)$ can be rewritten as

$$y - E(y) = (y - E(y|\boldsymbol{x})) + (E(y|\boldsymbol{x}) - E(y))$$

HINT 3: Let $u = y - E(y|\boldsymbol{x})$. Then $E[g(\boldsymbol{x})u] = 0$ for any function $g(\boldsymbol{x})$ (see part a of this exercise).

ANSWERS exercise 2

2a)

$$\begin{aligned} E[g(\boldsymbol{x})u] = E(E[g(\boldsymbol{x})u|\boldsymbol{x}]) = E(g(\boldsymbol{x})E[u|\boldsymbol{x}]) = E(g(\boldsymbol{x})E[y - E[y|\boldsymbol{x}]|\boldsymbol{x}]) = \\ E(g(\boldsymbol{x})(E[y|\boldsymbol{x}] - E[y|\boldsymbol{x}]) = E(g(\boldsymbol{x}) * 0) = 0 \end{aligned}$$

2b) Let $u = y - E(y|\boldsymbol{x})$ and $g(\boldsymbol{x}) = E(y|x) - E(y)$

$$\begin{aligned} var(y) &= E[(y - E(y))^2] \\ &= E[((y - E(y|\boldsymbol{x})) + (E(y|\boldsymbol{x}) - E(y)))^2] \\ &= E[(u + g(\boldsymbol{x}))^2] \\ &= E[u^2] + E[g(\boldsymbol{x})^2] + 2E[g(\boldsymbol{x})u] = \\ &= E[u^2] + E[g(\boldsymbol{x})^2] = \\ &= E[(y - E(y|x))^2] + E[(E(y|x) - E(y))^2] \\ &= E\{E[(y - E(y|x))^2|\boldsymbol{x}]\} + E[(E(y|x) - E(y))^2] \\ &= E[var(y|\boldsymbol{x})] + var(E(y|\boldsymbol{x})) \end{aligned}$$

In the third line we use hint 3

$$E[g(\boldsymbol{x})u](= E[(E(y|\boldsymbol{x}) - E(y))(y - E(y|\boldsymbol{x}))]) = 0$$

3. 14 points

Consider the simple regression model with one nonconstant explanatory variable z

$$y_i = \beta_1 + \beta_2 z_i + \varepsilon_i \tag{13}$$

and let x_i be a *binary* (dummy) instrumental variable for z (i.e. x_i can take on only two values 0 and 1). Rewriting model (13) yields:

$$y_i \underset{5}{=} \tilde{\boldsymbol{z}}_i'\boldsymbol{\beta} + \varepsilon_i \tag{14}$$

where $\tilde{\boldsymbol{z}}_i = (1, z_i)'$ and $\boldsymbol{\beta} = (\beta_1, \beta_2)'$. Since the model is just identified, one can use the following IV estimator to estimate $\boldsymbol{\beta}$:

$$\hat{\boldsymbol{\beta}} = \left(\sum_{i=1}^{n} \tilde{x}_i \tilde{z}_i' \right)^{-1} \sum_{i=1}^{n} \tilde{x}_i y_i \tag{15}$$

where $\tilde{\boldsymbol{x}}_i = (1, x_i)'$ and $\hat{\boldsymbol{\beta}} = (\hat{\beta}_1, \hat{\beta}_2)'$.

(a) Use equation (15) to show that

$$\hat{\beta}_2 = \left(\sum_{i=1}^{n} (x_i - \bar{x})(z_i - \bar{z}) \right)^{-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \tag{16}$$

ANSWER

Equation (15) implies that

$$\begin{pmatrix} n & \sum_{i=1}^{n} z_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i z_i \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{pmatrix}$$

or

$$\hat{\beta}_1 = \bar{y} - \bar{z}\hat{\beta}_2 \tag{17a}$$

$$\left(\sum_{i=1}^{n} x_i \right) \hat{\beta}_1 + \left(\sum_{i=1}^{n} x_i z_i \right) \hat{\beta}_2 = \sum_{i=1}^{n} x_i y_i \tag{17b}$$

Substitution of equation (17a) onto (17b) and rearranging yields

$$\left(\sum_{i=1}^{n} x_i (z_i - \bar{z}) \right) \hat{\beta}_2 = \sum_{i=1}^{n} x_i (y_i - \bar{y}) \tag{18}$$

Since $\sum_{i=1}^{n} \bar{x}(z_i - \bar{z}) = \sum_{i=1}^{n} \bar{x}(y_i - \bar{y}) = 0$, equation (18) can be rewritten as follows:

$$\left(\sum_{i=1}^{n} (x_i - \bar{x})(z_i - \bar{z}) \right) \hat{\beta}_2 = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \tag{19}$$

or

$$\hat{\beta}_2 = \left(\sum_{i=1}^{n} (x_i - \bar{x})(z_i - \bar{z}) \right)^{-1} \left(\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \right) \tag{20}$$

q.e.d.

(b) Use formula (16) to show that

$$\hat{\beta}_2 = \frac{(\bar{y}_1 - \bar{y}_0)}{(\bar{z}_1 - \bar{z}_0)} \tag{21}$$

where $\bar{y}_1$ and $\bar{z}_1$ are the sample averages of y_i and z_i over the part of the sample with $x_i = 1$, and where $\bar{y}_0$ and $\bar{z}_0$ are the sample averages of y_i and z_i over the part of the sample with $x_i = 0$. This estimator, known as the *grouping estimator*, was first suggested by Wald (1940).

ANSWER Let the set $I_1 = \{i|x_i = 1\}$ contain n_1 element (and the set $I_0 = \{i|x_i = 0\}$ n_0 elements. Given the binary nature of variable x_i, equation (18 can be rewritten as follows:

$$
\begin{aligned}
\hat{\beta}_2 &= \left(\sum_{i=1}^{n} x_i(z_i - \bar{z})\right)^{-1} \left(\sum_{i=1}^{n} x_i(y_i - \bar{y})\right) \\
&= \left(\sum_{i\epsilon I_1} z_i - n_1\bar{z})\right)^{-1} \left(\sum_{i\epsilon I_1} y_i - n_1\bar{y})\right) \\
&= (\bar{z}_1 - \bar{z}))^{-1} (\bar{y}_1 - \bar{y})) \\
&= \left(\bar{z}_1 - \left(\frac{n_1}{n}\bar{z_1} + \left(1 - \frac{n_1}{n}\right)\bar{z_0}\right)\right)^{-1} \left(\bar{y}_1 - \left(\frac{n_1}{n}\bar{y_1} + \left(1 - \frac{n_1}{n}\right)\bar{y_0}\right)\right) \\
&= \left(\left(1 - \left(\frac{n_1}{n}\right)\right)(\bar{z}_1 - \bar{z_0})\right)^{-1} \left(\left(1 - \left(\frac{n_1}{n}\right)\right)(\bar{y}_1 - \bar{y_0})\right) \\
&= (\bar{z}_1 - \bar{z_0})^{-1} (\bar{y}_1 - \bar{y_0})
\end{aligned}
$$

Table 1: Exercise 1, regression results

	equation 1	equation 2	equation male	equation female	aux. regression 1	2sls	first stage regression	aux. regression 2	aux. regression 3
Dependent variable	lwage	lwage	lwage	lwage	e^2_{lwage}*	lwage	educ	lwage	$\hat{u}_{lwage}$***
VARIABLES									
educ	0.383	0.398	0.391	0.353	0.0540	0.162		0.162	
	(0.0332)	(0.0338)	(0.0431)	(0.0517)	(0.0250)	(0.0220)		(0.0217)	
educ2	-0.00980	-0.0101	-0.0104	-0.00830	-0.00186				
	(0.00118)	(0.00121)	(0.00153)	(0.00185)	(0.000894)				
exper	0.0507	0.0520	0.0505	0.0492	0.000472	0.0680	-0.360	0.0680	-1.96e-05
	(0.00207)	(0.00211)	(0.00478)	(0.00241)	(0.00156)	(0.00821)	(0.00855)	(0.00809)	(0.00182)
ability	0.0271	0.0257	0.0253	0.0294	0.00438				
	(0.00467)	(0.00477)	(0.00636)	(0.00686)	(0.00353)				
math	0.00372	0.00502	0.0164	-0.00998	-0.00664				
	(0.00758)	(0.00773)	(0.0104)	(0.0110)	(0.00572)				
SE	0.155		0.197	0.108	0.0415	0.141	0.405	0.141	3.64e-05
	(0.0130)		(0.0173)	(0.0196)	(0.00984)	(0.0164)	(0.0634)	(0.0162)	(0.0135)
SW	0.219		0.226	0.213	0.00760	0.170	1.133	0.170	0.000634
	(0.0236)		(0.0315)	(0.0356)	(0.0179)	(0.0346)	(0.114)	(0.0341)	(0.0243)
female	-0.256	-0.255			0.0165	-0.203	-0.882	-0.203	0.000400
	(0.0121)	(0.0123)			(0.00913)	(0.0230)	(0.0563)	(0.0227)	(0.0120)
payrsed							0.0556		0.00288
							(0.0115)		(0.00245)
mayrsed							0.00604		-0.00355
							(0.0123)		(0.00262)
$\hat{v}$**								-0.0372	
								(0.0219)	
Constant	-2.130	-2.233	-2.156	-2.185	-0.269	-1.005	17.06	-1.005	0.00618
	(0.225)	(0.229)	(0.311)	(0.346)	(0.170)	(0.389)	(0.149)	(0.383)	(0.0317)
Observations	4,744	4,744	2,490	2,254	4,744	4,744	4,744	4,744	4,744
R^2	0.439	0.415	0.289	0.398	0.007	0.393	0.307	0.412	0.000
SSR	643.600	670.795	315.191	323.880	367.285	695.766	15366.676	674.546	695.495
F test exclusion restrictions							46.736		

Standard errors in parentheses

* e_{lwage} denotes the residual of the regression 1 (cf. column 'equation 1')

** $\hat{v}$ denotes the residual of the first stage regression (cf. column 'first stage reg.')

*** $\hat{u}_{lwage}$ denotes the 2sls residual of equation (4) (cf. column '2sls')